GG-S-1340A
November 24, 1975
SUPERSEDING
Int. Fed. Spec. GG-S-001340
December 15, 1969 and
Fed. Spec. GG-S-751a (In part)
November 9, 1940

FEDERAL SPECIFICATION
STERILIZER, SURGICAL INSTRUMENT AND SUPPLY GRAVITY AIR REMOVAL, NON-PORTABLE (HEAT AND MOISTURE STABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The sterilizer covered by this specification shall be the horizontal, vacuum-pressure, gravity air removal type for processing heat- and moisture-stable materials, using saturated steam as the sterilizing agent.

1.2 Classification. Sterilizer shall be the following types, sizes, and styles as specified (see 6.2):

1.2.1 Type.

Type I - General purpose, steam powered; for heating with steam from a building supply line.

Type II - General purpose, electrically powered; for heating with steam from an integral, electric steam generator (size 1 and 2 only).

Type III - Laboratory, steam powered; for heating with steam from a building supply line.

Type IV - Laboratory, electrically powered; for heating with steam from an integral, electric steam generator (size 1 and 2 only).

1.2.2 Size. The minimum clearance dimensions of the chamber side of the inner shell excluding space for baffle plate (see 3.6.1.1) shall be as shown in table I.

TABLE I Inner shell dimensions (inches)

Size	Width	Height	Length
1	16	16	26
2	20	20	38
3	24	36	36
4	24	36	48
5	24	36	60

1.2.3 Style.

Style A - Single door, cabinet enclosed (free standing).
Style B - Single door; for recessing through one wall partition.
Style C - Double door; for recessing through one wall partition (for goods pass-thru).
Style D - Double door (sizes 4 & 5); for recessing through two wall partitions (for goods pass-thru).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6530

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Federal Specifications:

CC-M-636 - Motor (Alternating Current), Fractional Horsepower.
DD-B-592 - Bottle, Sterile Fluid Storage.
FF-C-77 - Casters, Institutional Duty.
QQ-N-281 - Nickel-Copper Alloy Bars, Plates, Rods, Sheets, Strips, Wire Forgings and Structural and Special Shaped Sections.
QQ-S-766 - Steel Plates, Sheets and Strip-Corrosion-Resisting.
PPP-T-60 - Tape, Pressure Sensitive Adhesive, Waterproof for Packaging.
PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843 - Cushioning Material, Cellulosic.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-C-132 - Crates, Open Wood; Maximum Capacity 2,500 pounds.
MIL-C-3774 - Crates, Wood, Open 12,000 and 16,000 pound capacity.
MIL-I-45208A - Inspection Requirements.
MIL-L-10547 - Sheet, Overwrap; Water Vapor Proof or Water Proof, Flexible.
MIL-P-116 - Preservation, Methods of.
MIL-S-36586 - Sterilization Test Strip, Bacterial Spore.

Military Standards:

MIL-STD-105 - Sampling Procedures, and Tables for Inspection by Attributes.
MIL-S-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society of Mechanical Engineers (ASME):

Boiler and Pressure Vessel Code, Section 1, Part PMB and Section VIII Pressure Vessels, Division 1.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 E. 47th St., New York, NY 10017.)

American National Standards Institute (ANSI) Standard

A 40.8 - National Plumbing Code

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

B 88 - Seamless Copper Water Tube.
B 135 - Seamless Brass Tube.
B 43 - Seamless Red Brass Pipe.
B 456 - Specification for Electrodeposited Coating of Nickel plus chromium.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Fire Protection Association (NFPA) Publication:

No. 70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch St., Boston, MA 02110.)

United States Pharmacopoeial Convention, Incorporated:

Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Compliance with standards.

3.1.1 Pressure vessel. The design, material and testing of each sterilizer chamber assembly shall conform to applicable requirements of the ASME Boiler and Pressure Vessel Code, Section VIII, Pressure Vessels, Division 1. In addition, the electric steam generator of a type II or type IV sterilizer shall conform to applicable requirements of Section 1, Part PMB of the referenced ASME Code.

3.1.1.1 Certification. The manufacturer shall furnish with each sterilizer certification as required by the referenced ASME Boiler and Pressure Vessel Code.

3.1.2 Electrical components. The electrical components of each sterilizer shall conform to applicable requirements of NFPA No. 70.

3.1.2.1 Motors. Shall be of sufficient size for the duty to be performed, and shall not exceed their nameplate ratings when the driven equipment is operating at specified capacity under the most severe conditions likely to be encountered. The motors shall have over-current protection.

3.1.2.1.1 Fractional horsepower. Shall be continuous duty, class A or B insulation, CC-M-636.

3.2 Materials.

3.2.1 Corrosion-resisting metal. Whenever corrosion-resisting metal is required, it shall be either nickel-copper alloy or corrosion-resisting steel.

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3.2.2 Corrosion-resisting steel. Corrosion-resisting steel shall conform to QQ-S-766, class 304 where welding is required; class 302, 304, 201, or 202, if formed without welding.

3.2.3 Nickel-copper alloy. Nickel-copper alloy shall conform to QQ-N-281.

3.2.4 Clad steel. Clad steel shall consist of nickel or nickel-copper alloy and steel, mill rolled under heat and pressure until they are integrally bonded over their entire interface.

3.2.5 Chromium plated metals. Chrome plate on die castings shall meet the requirements for service condition SCL in ASTM B 456 except equivalent nickel thickness shall be 0.3 Mil. Chromium plate on copper alloys shall also meet SCL except the equivalent nickel thickness shall be 0.1 Mil.

3.2.6 Fasteners. Where brass, bronze, copper, or corrosion-resisting-metal parts are welded or otherwise joined to each other, the welding rods or other fasteners shall, unless otherwise specified herein, be of the same material as that so joined. Where necessary to join these parts to dissimilar metals, the welding rods or other fasteners shall be of the same material as the part welded or fastened to the dissimilar metal. Tetrafluoroethylene tape or other suitable inert compound shall be applied to threaded connections involving dissimilar metals.

3.2.7 Steam, water and waste lines. Install these lines on the sterilizer to conform to the requirements of ANSI A 40.8. Unless otherwise specified (3.6.5.2.2 and 3.6.5.2.3), the lines shall consist of pipe or tubing which conforms to the ASTM designations specified below; line connectors (excluding accouterments such as valves, strainers and regulators) must be brass or copper.

Pipe:	Brass	ASTM B 43
Tubing:	Copper	ASTM B 88
	Red Brass	ASTM B 135, Alloy 1

3.2.8 Manual valves. Hand valves on steam and water supply lines shall be bronze with synthetic discs or equivalent parts. Hand valves shut off requiring routine operation shall have renewable nickel copper alloy seats. Each valve shall withstand a 350-psig hydrostatic test or be leakproof when tested at 100-psig air pressure with valve body submerged in water. Each valve shall have a renewable low heat conducting and non-loosening handle. Durable letters on the handle shall show the purpose of the valve. All hand valves shall be easily accessible from the control end of the sterilizer.

3.3 Design. The sterilizer shall operate on the gravity-air-removal principle and have horizontal, steam-jacketed chamber and steamtight door or doors, as specified. The sterilizer shall be provided with operating controls, exhaust system, chamber drain system, heating system, mounting device, and materials-handling accessories (see 3.6.8).

3.4 Performance characteristics. Operation of the sterilizer shall comprise the following cycle-phase sequences:

3.4.1 Admission of steam to chamber. Steam shall be admitted to the sterilizing chamber to achieve the temperatures specified below. During the exposure period at selected chamber temperature, chamber pressure shall be approximately the same as jacket pressure. The times required to reach chamber temperatures shall not exceed those specified in 4.2.3.3.1:

Types I and II	- Either 250°F (121-123°C) or 270° - 274°F (131-134°C), as selected by the operator.
Types III and IV (without low-temperature control)	- Any temperature between 212° and 274°F. (100° and 134°C), as selected by the operator.
Types III and IV (with low-temperature control; see 3.6.3.1.3)	- Any temperature between 172° and 274°F. (78° and 134°C), as selected by the operator.

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3.4.2 Exposure period. When the selected temperature is reached, it shall be maintained automatically throughout the selected exposure period to sterilize the surgical-goods test pack as specified in 4.2.3.3.3 through 4.2.3.3.3.2.

3.4.3 Exhausting steam from chamber. Each sterilizer shall provide either fast or slow exhaust of steam from the chamber, as selected by the operator, following the exposure period. The exhaust system shall function to conform to the requirements of 4.2.3.3.2.1 and 4.2.3.3.2.2.

3.4.4 Drying. Following exhaust of steam from the chamber, wrapped goods shall be dried to conform to the requirements of 4.2.3.3.3.2.

3.5 Maintainability.

3.5.1 Service access. The electric steam generator heaters, gauges, and pressure control shall be behind a panel at the front (control end) of a type II or type IV. The panel shall open or be removable without tools and shall provide access for viewing and servicing the components behind it.

3.5.2 Wiring diagrams. Shall be securely attached to sterilizer in accessible location such as inside of a control panel, terminal board cover or control housing door, or furnished separately in protective covers.

3.5.3 Instruction books and parts list. Unless otherwise specified (see 6.2), two copies of booklets containing the following information shall be furnished with each sterilizer:

(a) Instruction for installing, operating and performing preventive maintenance on the equipment.

(b) List of service parts (identified by manufacturer's part number) and quantity required for preventive maintenance.

(c) When specified (see 6.2), additional information such as roughing-in drawings, power requirements, structural modifications, and special support devices shall be furnished.

3.5.3.1 Distribution. Unless otherwise specified (see 6.2), the instruction books and parts lists shall be shipped with the sterilizer to which they pertain.

3.5.3.2 Approval of data. Unless otherwise specified (see 6.2), two copies of instruction books and parts list shall be forwarded to the Procuring Agency for review and approval prior to delivery of the sterilizer. The data is to be furnished sufficiently early to allow time for revision of data as required by the Procuring Agency as a condition of approval.

3.5.3.3 Additional data. When specified (see 6.2), fifteen copies of instruction books and parts lists shall be forwarded to the Procuring Agency upon completion of the contract or purchase order.

3.6 Components.

3.6.1 Chamber assembly. This assembly shall be of welded construction and consist of inner and outer shells placed one within the other to provide a steam space (steam jacket) over the lengthwise exterior surfaces of the inner shell. The shells for a single-door sterilizer shall be joined by a door frame on one end and a backhead on the opposite end. The shells for a double-door sterilizer shall be joined at each end by a door frame. The space between the two shells of a size 3, 4, or 5 may be sealed by a frame closure and separate end ring.

3.6.1.1 Inner shell. The inner shell shall be hot-rolled carbon steel. The inside surfaces shall be clad with nickel or nickel-copper alloy as specified in 3.2.4. The shell for a size 3, 4, or 5 shall include nickel-copper-alloy loading car tracks, secured to the chamber without exposing the ferrous metal beneath the cladding to the sterilant.

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3.6.1.2 Outer shell. The outer shell shall be hot-rolled carbon steel. The exterior (including backhead, if shell is for a single-door sterilizer) shall be insulated with a glass-fiber blanket commercially rated one inch thick; exterior surfaces of glass-fiber shall be covered with aluminum foil. The insulation shall be held on the sterilizer with thermosetting tape or aluminum straps.

3.6.1.3 Backhead. The backhead for a single-door sterilizer shall be hot-rolled carbon steel. Surfaces exposed to the sterilizing chamber shall be clad as specified in 3.2.4.

3.6.1.4 Baffle plate. A nickel-copper-alloy baffle shall shield the steam opening inside the chamber.

3.6.1.5 Filter. The filter supplied with the drying system shall admit filtered air to the chamber and entrain vapors and odors to be conducted to exhaust. Filter shall be the ceramic-labyrinth type with maximum pore size of 50 microns or a cast column containing at least 1/2 ounce of tightly packed, nickel-copper-alloy wool, and equipped with a condensate-drip device. The filter shall be sterilized with each cycle.

3.6.1.6 Door Frame. Each door frame, frame closure or end ring shall have a smoothly ground, nickel-copper-alloy surface for door gasket contact.

3.6.2 Door. Size 1 or 2 shall have manually operated door(s). Sizes 3, 4, and 5 shall have power driven or mannually operated door(s) as specified (see 6.2). Each door shall be attached by a bearing-mounted hinge. The hinge shall be adjustable for accurate, door-gasket alignment and easy door opening and closing. The door swing shall allow unobstructed access to the chamber. Door swing shall be right or left hand as specified (see 6.2). Door swing for a size 1 or 2 shall be reversible in the field without parts, welding or drilling. The opening and closing mechanism shall be so designed that the holding members will be fully engaged before gasket seal can be established.

3.6.2.1 Door material. The material for the door shall be cast bronze, hot-rolled carbon steel clad on the chamber side with nickel or nickel-copper-alloy as specified in 3.2.4, or corrosion-resisting steel as specified in 3.2.2. The hinge shall be bronze, brass, corrosion-resisting steel, or chromium-plated steel. The chamber side shall be provided with a gasket warranted by the manufacturer to be suitable for at least six months after initial operation. The gasket shall be held by a groove in the door or by corrosion-resisting retainers on the door. The groove in a clad door shall be fabricated to prevent exposure of the ferrous metal under the cladding to the sterilant. Outside surfaces of the door plate and door-securing members shall be concealed (to the maximum permitted by the ASME Boiler and Pressure Vessel Code) by a cover. The cover shall be of the same material as specified for the sterilizer panels (see 3.6.7). Insulation shall be provided between the door and door-cover panel.

3.6.2.2 Safety lock. Each door shall lock automatically when pressure in the chamber rises to between 0 and 4 psig. The lock shall not release until the pressure is reduced to approximately atmospheric. For a power-operated door, the electrical control system shall be so designed that power to the door shall terminate when the chamber is pressurized. Components of the safety lock exposed to steam shall be brass, bronze or corrosion-resisting steel.

3.6.2.3 Manually-operated door. All functions of opening and closing a manually-operated door and positioning, tightening and loosening the door-holding members shall be by a low-heat-conducting handwheel or a quick-throw arm and low-heat-conducting handwheel. It shall be possible to tighten, but not loosen, the door while the chamber is under pressure. In lieu of the ability to tighten the door with the chamber pressurized, the door may be equipped with an audible indication when seal is effected and a device to prevent overtightening the door.

3.6.2.4 Power-operated door. A power-operated door shall be equipped with a mechanism to close and lock the door pressure tight, close it without compressing the gasket, and unlock and open it fully. The drive motor may be the intermittent-duty type. The door shall also be manually operable. The power-drive controls shall include door-positioning selectors and automatic overload protection. The door shall be so designed that it will stop automatically if it encounters an obstruction while being opened or closed. Door-positioning controls requiring operator attention shall be conveniently accessible. Conduit, gears, wiring and other components of the door cover or front panel (see 3.6.7.1). Maintenance of the door and its removal from the sterilizer shall not require disconnecting any soldered wire joints. The door operating system shall be fused for operation on 120 volt, 60-Hz electric power.

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3.6.3 Operating controls. The sterilizer shall be equipped with either manual or automatic controls as specified (see 6.2).

3.6.3.1 Automatic control. Shall be a motor-driven mechanism to control all functions of the sterilizer. The control shall also include, and operate in conjunction with, associated instrumentation; a temperature indicator-recorder-controller; and supplemental, low-temperature system, if specified for type III or type IV. Push-button selectors shall be provided to offer liquids, wrapped goods, and unwrapped goods cycles. After closing the door(s) and selecting the sterilizing temperature, exposure time and type of load, no further operator attention shall be necessary until completion of the cycle is indicated by an alarm and signal light. The audible alarm shall cease automatically when the door(s) of the sterilizer is opened, or within approximately two minutes after completion of the cycle, whichever occurs first. Sequential signal lights shall indicate each phase of the sterilization cycle. The control shall be so designed that a cycle will start automatically only when the door(s) is (are) closed and the holding members engaged. Except for a low-temperature system, the control unit shall also be manually operable without electrical power, by a single programming wheel, dial or handle to select, bypass or extend any phase of a processing cycle. The control shall be driven by an intermittent-duty motor. The program selector buttons shall be so designed that, when depressed, alternate cycle selections cannot be made without first depressing a reset button or waiting until the selected cycle has been completed.

3.6.3.1.1 Associated instrumentation. Shall consist of at least the following items:

(a) Means of obtaining electrical power to unit.
(b) Cycle phase lights.
(c) Chamber-vacuum pressure gauge.
(d) Jacket-pressure gauge.
(e) Sterilizing exposure timer (Adjustable for any interval between 0 and 90 minutes).
(f) Drying phase timer (for fixed or variable timing).
(g) On opposite end from main control panel for style C and D sterilizers

(1) Chamber vacuum pressure gauge.
(2) Light (completion of sterilizing cycle).
(3) Audible signal (completion of sterilizing cycle).
(4) Single alarm device (style C and D), provided it is clearly audible to both ends).

3.6.3.1.1.1 Main switch. A means shall be provided for de-energizing the main control.

3.6.3.1.1.2 Cycle phase lights. Lights shall be provided to indicate each phase of the sterilizing cycle.

3.6.3.1.1.3 Chamber-vacuum pressure gauge. Shall be provided and shall be accurate within limits specified in 4.2.3.3.5.

3.6.3.1.1.4 Jacket pressure gauge. Shall be provided with accuracy as specified in 4.2.3.3.5.

3.6.3.1.1.5 Sterilizing exposure timer and indicating light. The sterlizing exposure timer provided shall be adjustable for any interval between 0 and 90 minutes and shall automatically reset upon completion of the selected time interval and if electric power fails. The timer and indicating light shall not actuate until the chamber temperature has reached the selected time for the cycle and shall also automatically reset if chamber temperature falls 2°F below sterilizing value.

3.6.3.1.1.6 Drying phase timer. The drying phase timer provided shall be (fixed or variable) and may be integral with the sterilizing exposure timer. The drying timer shall automatically reset upon completion of the selected time interval and if electric power fails.

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3.6.3.1.1.7 Associated instrumentation for the sterilizer end opposite the main control panel. Style C or style D double-door sterilizers shall be provided with the following controls on the opposite end from the main control panel.

(1) Chamber vacuum pressure gauge (accuracy of gauge shall be as specified in 4.2.3.5.5).
(2) Light indicating completion of sterilizing cycle.
(3) Audible signal to indicate completion of sterilizing cycle or a single alarm device indicating completion of cycle may be supplied, provided it is clearly audible at both ends of sterilizer.

3.6.3.1.2 Temperature indicator-recorder-controller. The indicator-recorder-controller shall control the exposure timer and indicate and record chamber temperature throughout the processing cycles with the accuracy specified in 4.2.3.3.5. This instrument on type III or type IV shall also provide selection and control of any processing temperature within the range of 212° and 274°F (100 and 134°C) without adjustment of the steam-supply-line pressure regulator. The recording chart mechanism shall be driven by a synchronous, 24-hour timing motor. The ink supply for the recorder shall be from a cartridge on the pen arm or by a capillary device. The cartridge or capillary supply shall be sufficient for at least three months use, and the ink supply shall be visible. One-hundred recording charts and a one-year ink supply shall be furnished. The temperature sensor shall be in the chamber drain line (see 3.6.6).

3.6.3.1.3 Supplemental, low-temperature system. When specified (see 6.2), supplemental low-temperature system (for a single door type III or type IV sterilizer) shall permit fully automatic processing of heat-coagulable and heat-sensitive materials at any temperature between 78° and 104°C (172° and 119°F) as selected on the indicator-recorder-controller. These temperatures shall be maintained accurate within the limits specified in 4.2.3.3.4.2. The low-temperature processing cycle shall be actuated by a separate push button or switch.

3.6.3.1.4 Location of automatic control components. Associated instruments and other components of the automatic control which require operator attention shall be in a panel above the door frame on the control end of the sterilizer. The additional chamber-vacuum-pressure gauge and in-cycle indicator light for a style C or style D sterilizer shall be above the door frame opposite the control end.

3.6.3.1.5 Interlock for style C and D. When specified (see 6.2 and 6.3), the sterilizer shall be so equipped that under routine operating conditions only one of the doors can be opened at a time and then only if the opposite door is closed pressure-tight. In addition, after starting a sterilizing-process cycle, the interlock shall prevent the door on the sterilized-goods-unloading end of the sterilizer from being opened until the cycle has been completed. The interlock shall include a bypass, operable by a removable key, to allow simultaneous opening of the doors for maintenance of the sterilizer. Signal lights shall also be provided as follows: at each end of the sterilizer, to indicate when the door at the opposite end is open or closed pressure-tight; and, on the end opposite the main operating controls, to indicate process-cycle completion.

3.6.3.2 Manual control. A manual control shall consist of a single programming wheel, dial or handle to control the processing-cycle-phase sequence, a temperature indicator-recorder, and associated instruments.

3.6.3.2.1 Temperature-indicator-recorder. The temperature-indicator-recorder shall operate on 120-volt, 60-Hz electric power and function automatically during the processing cycles. Accuracy of the instrument shall be as specified in 4.2.3.3.5. The recording chart mechanism shall be driven by a synchronous, 24-hour timing motor. The sensor shall be in the chamber drain line. The ink supply for the recorder shall be by a cartridge on the pen arm or by a capillary device. The cartridge or capillary shall be sufficient for at least three months use and the ink supply shall be visible. One-hundred recording charts and a one-year supply of ink shall be provided.

3.6.3.2.2 Associated instruments. Associated instruments shall include a chamber-vacuum-pressure gauge (two, if a double-door sterilizer), jacket-pressure gauge, and a timer adjustable between 0 and 60 minutes. These instruments shall be accurate within the limits specified in 4.2.3.3.5.

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3.6.3.2.3 Location of manual control components. The cycle-programming wheel, dial or handle, temperature-indicator-recorder and associated instruments shall be in a panel above the door frame on the control end of the sterilizer. The additional chamber vacuum-pressure gauge for a style C or style D sterilizer shall be above the door opposite the control end.

3.6.4 Exhaust system. The exhaust system shall receive and facilitate the disposal of steam and condensate from the sterilizer chamber. The system shall include a condenser assembly or atmospheric exhaust, as specified (see 6.2), and be designed to prevent backflow of steam and condensate into the sterilizer or building-water supply system.

3.6.4.1 Condenser assembly. A condenser assembly shall consist of a corrosion-resisting-steel, copper or brass heat exchanger, waste line, and building waste-line funnel. Steam and condensate shall enter the heat exchanger where the steam is condensed. Condensate shall then flow from the heat exchanger, through the waste line, and into the funnel. The funnel shall be sized or designed to prevent spillage from the top. The waste line shall terminate at one side or rear of the sterilizer. The condenser assembly on a sterilizer with automatic controls shall function through those controls, and shall also be manually operable from the control end. On an automatically controlled sterilizer, the flow of cooling water through the condenser shall be automatically regulated so that usage of water is held to a minimum consistent with the amount of steam it must condense during each phase of the sterilization cycle. The condenser assembly on a sterilizer with manual controls shall be operable by a handvalve at the control end of the sterilizer or may be operable through the manual control (see 3.6.3.2).

3.6.4.2 Atmospheric exhaust. This shall consist of an exhaust line with condensate drain and a building waste line funnel. The funnel shall be sized or designed to prevent spillage of condensate from the top. The exhaust line shall terminate at the rear or one side of the sterilizer in a fitting for ready connection to a building vent stack.

3.6.4.3 Biological seal. When specified (see 6.2 and 6.3), a style C or D sterilizer shall be equipped with a seal which shall prevent the passage of air-borne microorganisms through the wall opening into which the designated end of the sterilizer is to be installed. The seal shall include a flange continuously welded to and completely around the exterior of the sterilizer outer shell; a mating flange welded to a channel which shall be sized to the thickness of the partition wall into which the affected end of the sterilizer is to be installed; and a gasket which shall wholly interconnect the two flanges. The gasket shall be secured to the flanges with metal clamping bars, bolts and washers. A single, continuous bar shall be used to hold the gasket to any given length or height dimension of each flange; bars shall be half-lapped and bolted at corner points. The flanges shall be carbon steel; the gasket, synthetic rubber. If necessary to extend any pipes, wires, valve stems or other such fittings through either flange or the gasket, the penetration points shall be sealed to provide the same degree of tightness as the flange or gasket through which such components may extend.

3.6.5 Heating system.

3.6.5.1 Types I and III. A type I or III shall operate on steam from an independent source delivered to the sterilizer at 50 to 80 psig dynamic. Sterilizer shall be provided with a steam-supply line to the jacket and a fitting for connection to a condensate return line or, if specified (see 6.2) a condensate line to waste. Condensate lines shall include a check valve and thermostatic trap. All lines shall terminate in fittings at the side or rear of the sterilizer for ready connection to building service lines. The steam-supply line shall include a strainer, a shutoff and a regulator that will automatically maintain the selected jacket pressures. The pressure regulator shall shall be adjustable over the range of 6 to 32 psig. The steam trap shall have a replaceable element and seat.

3.6.5.2 Types II and IV. A type II or type IV (size 1 or size 2) shall be provided with an integral, electrically powered steam generator. The generator assembly shall be beneath, and not protrude beyond, the sterilizing chamber assembly.

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3.6.5.2.1 Operation. The generator shall be automatically controlled to generate and maintain a constant supply of steam to the sterilizer pressure regulator at 50 to 80 psig. The generator shall include water and waste systems, heaters, and insulation. Valves, gauges and switches requiring operator attention shall be accessible from the control end of the sterilizer.

3.6.5.2.2 Water and waste systems. The generator shall be for operation on water-supply-line pressure of 20 to 50 psig and of the automatic, water-filling type. The generator shall be equipped with a turbine-type water pump powered by a continuous-duty motor. Water and waste lines on the generator shall terminate in fittings at the rear or side, ready for connection to building service lines. The water line shall include a strainer, vacuum breaker and handvalve shutoff. The waste line shall include a handvalve shutoff. Water and waste lines may be black iron. The generator shall be so arranged that it can be easily flushed without disassembly.

3.6.5.2.3 Interconnecting piping. Shall be installed between the generator output port and the sterilizer jacket. The steam supply line shall be equipped with a strainer, a handvalve shutoff and a pressure regulator that will automatically maintain the selected pressure in the steam jacket. The regulator shall be adjustable over the range of 60 to 32 psig. Steam lines between generator and sterilizer jacket may be black iron.

3.6.5.2.4 Heaters. Total nominal capacity of the heaters shall be adequate to meet performance requirements of 4.2.3.3.1. The heater wires shall be 80 percent nickel and 20 percent chromium alloy enclosed in magnesium-oxide-compacted, corrosion-resisting-steel or copper sheaths. Automatic low-water and thermal protection shall be provided for the heaters. The heaters shall operate on 3-phase, 60-Hz, electric-service voltage as specified (see 6.2). An off-on line switch shall be provided.

3.6.5.2.5 Insulation. The body of each generator shall be insulated with material conforming to that specified in 3.6.1.2.

3.6.5.3 Safety-valve setting. Safety valve for the sterilizer jacket and chamber shall be set to relieve pressure in excess of the rated maximum operating pressure of the vessel and be sealed to prevent change of adjustment.

3.6.6 Chamber-drain system. Each sterilizer shall be provided with an opening in the chamber floor (at approximately the front center) to pass air, steam and condensate during the processing cycles. The chamber floor shall be designed to prevent water from collecting and drain fitting shall be flush with the chamber floor. The opening shall be fitted with a removable, brass, bronze or corrosion-resisting-metal strainer to catch lint and dirt. A drain line shall be provided and installed between the opening and the sterilizer exhaust system. The line shall include a thermostatic steam trap and a check valve to prevent backflow of steam and condensate into the chamber. The line for a type III or type IV sterilizer shall, in addition, be equipped with a hand-valve (operable from the control end of the sterilizer) to bypass the thermostatic trap.

3.6.7 Mounting devices. The sterilizing chamber assembly shall be supported by a steel stand equipped with adjustable, corrosion-inhibiting floor flanges. The various styles of sterilizers (see 1.2.3) shall be fitted with corrosion-resisting steel panel(s) as shown below:

Style A A front panel, two side panels and a top panel to enclose the sterilizer body, pipes and fittings.

Style B A front panel.

Style C Front and back panels, two side panels and a top panel to enclose the sterilizer body, pipes, and fittings that project beyond the wall partition.

Style D Front and back panels.

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3.6.7.1 Front and back panels. Each front and back panel shall comprise a lower section that will cover the area to each side of and below the door frame and any area above the door frame not covered by an instrument panel. Front and back panels shall be not less than 0.050 inch thick and fitted with adjustable kickplates. No fasteners shall be exposed from the exterior of the sterilizer and the panel(s) shall include gaskets or other suitable means which will ensure a tight fit along the entire top and side edges where they abut wall partitions or companion top and side panels.

3.6.7.2 Side and top panels. Shall be formed from sheets not less than 0.050 inch thick. No fasteners shall be visible from the exterior of the sterilizer.

3.6.7.3 Canopy. When specified (see 6.2 and 6.4), a ventilating canopy shall be provided, for a style A or C sterilizer. The canopy shall replace the sterilizer top panel and shall be attached to the sterilizer frame in such manner that the canopy will snugly fit to the sterilizer cabinet panels and not interface with their ready removal. The canopy shall form a plenum which shall terminate at the top center (of the end nearest the wall into or against which the sterilizer is to be installed) in a flanged duct for ready connection to the building vent system. The plenum shall be at least 19 inches deep at its vent connection point and shall slope to overhang the sterilizer cabinet end panel approximately eight inches. Air-intake openings shall be provided in the canopy overhang and, if required, in the sterilizer side cabinet panels to meet the airflow requirements of 6.4. Such air-intake openings shall be the stamped-louver type and shall include adjustable, knob-and-spring-actuated dampers. The canopy shall be of welded construction and of the same material as the sterilizer cabinet panels. Weldments on surfaces of the canopy exposed to view from the front or sides of the sterilizer shall be ground and polished so that the canopy will appear as a seamless unit. If canopy is for a style A sterilizer, a corrosion-resisting-steel panel shall also be included to fully enclose the rear of the sterilizer cabinet.

3.6.8 Materials-handling accessories. When specified (see 6.2), the sterilizer shall be provided with one or more of the materials-handling accessories listed in table II (see 6.1.1).

TABLE II Accessories

Accessories	Type I or II Sizes					Type III or IV Sizes				
	1	2	3	4	5	1	2	3	4	5
Inst. trays	X									
Bottom shelf	X	X				X	X			
Two shelves	X	X				X	X			
Three shelves	X	X				X	X			
Loading car		X	X	X	X		X	X	X	X
Carriage		X	X	X	X		X	X	X	X

NOTE: "X" indicates the accessories normally specified for sterilizers

3.6.8.1 Instrument trays. Each instrument tray shall be made of welded, nickel-copper alloy. It shall have a wire-mesh or perforated bottom and a carrying handle at each end which will fold or retract approximately flush with the top or end of the tray. The tray shall be not less than 20-1/2 inches long by 10-1/2 inches wide by 2-1/2 inches deep.

3.6.8.2 Bottom shelf. The bottom shelf shall be made of nickel-copper alloy wire road and extend the full, usable length and width of the interior of the sterilizer chamber. The shelf shall be of welded construction and provided with nickel-copper alloy support to prevent goods on the shelf from contacting the chamber floor.

3.6.8.3 Shelves. Each shelf shall be made of nickel-copper-alloy wire rod and extend the full, usable length and width of the sterilizing chamber. When fully loaded, the shelf shall slide freely on nickel-copper-alloy angle guides. The angle guides, if affixed to the sterilizing chamber walls, shall be installed without exposing the ferrous metal beneath the chamber cladding to the sterilant. Stops shall be provided to allow each shelf to be withdrawn approximately one-half of its length.

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procedures for visual examination shall be

3.6.8.4 Loading cars.

3.6.8.4.1 Size 2. The loading car for size 2 shall consist of a framework, two shelves, axles, and wheels. Minimum outside dimensions of the framework shall be 33 inches long by 18 inches wide by 14 inches high. The upper shelf shall be adjustable to three heights. The loading car, while containing an evenly distributed, 200-pound load, shall roll freely on a removable, corrosion-resisting-metal, chamber-track assembly which shall be furnished. The wheels shall be bearing mounted and made of nickel-copper-alloy, bronze or brass. The axles shall be nickel-copper-alloy and other parts of the car shall be constructed of welded nickel-copper-alloy.

3.6.8.4.1.1 Sizes 3, 4 and 5. The loading car for a size 3, 4, or 5 shall be a welded, nickel-copper-alloy framework (with open sides) supported on corrosion-resisting-steel, bronze or brass, bearing-mounted wheels attached to nickel-copper-alloy axles. The framework shall contain a full-length, full-width bottom shelf, a removable load-retaining gate (to form one side closure) and not less than two adjustable, full-width, full-length or four adjustable half-width, full-length shelves. For size 5, eight adjustable half-width, half-length shelves may be furnished in lieu of the four half-width, full-length shelves. The adjustable shelves shall be removable and provide at least seven loading heights. A means shall be provided at both ends of the carriage to retain smaller packs from falling from the shelves during transportation to and from the sterilizer. The shelves and load-retaining gate shall be made of welded, nickel-copper-alloy wire rod. While containing an evenly distributed, 1,000 pound load, the car shall roll freely on the tracks inside the sterilizing chamber (see 3.6.1.1).

TABLE III Loading car dimensions

Size	Minimum outside dimensions (in.) of car framework		
	Length	Width	Height
3	35	21	28
4	47	21	28
5	59	21	28

3.6.8.5 Carriages.

3.6.8.5.1 Size 2. The carriage for size 2 shall be a table-like structure of corrosion-resisting steel, supported by four casters (2 swivel and 2 fixed) with non-marking polyurethane or synthetic rubber tires. The carriage shall include a top surface with height-adjustable tracks or wheels to receive the loading car from the sterilizer, a shelf beneath the track assembly for storing and transporting supplies, and a handle to facilitate maneuverability. Latches shall be provided to secure the carriage to the sterilizer while the loading car is wheeled onto the carriage tracks and to secure the loading car while it is being transported on the carriage. These latches shall be operable from the handle end of the carriage.

3.6.8.5.2 Sizes 3, 4 and 5. The loading car carriage for a size 3, 4 or 5 shall be a caster-mounted structure containing height-adjustable tracks or wheels to receive the loading car from the sterilizer. The casters shall conform to class C, style A, FF-C-77, except they shall be fitted with non-marking polyurethane or synthetic rubber tires. Two casters shall be fixed and two shall be swivel. The carriage shall be provided with a handle at the push end. Spring-loaded latches shall be provided to secure the carriage to the sterlizer while the loading car is wheeled onto the carriage tracks and to secure the loading car while it is transported on the carriage. These latches shall be operable from the handle end of the carriage. The carriage shall also be equipped with resilient bumpers or similar devices to protect walls or furniture from damage should the carriage strike such objects. The carriage structure shall be welded or bolted and of corrosion-resisting-steel or coated-steel construction.

3.6.9 Finish. The finish of the sterilizer and materials-handling accessories shall be free of burrs or roughness which could cause personal injury, impede cleaning or damage goods processed in the sterilizer. Unless otherwise specified herein, the finish of the various components of the sterilizer shall conform to the manufacturer's regularly employed commercial standards.

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3.6.9.1 Corrosion protection. Exposed surfaces of the outer shell, including backhead of a single-door sterilizer and sterlizer mounting stand, shall be suitably coated to protect the parts from corrosion.

3.6.9.2 Door parts. Corrosion-resisting metal exterior door and door-frame parts (except handwheel) not concealed by door cover, shall have a satin finish; parts of other than corrosion-resisting metal shall be chromium plated and then polished to a satin or brighter finish.

3.6.9.3 Carriage. A carriage not of corrosion-resisting-metal construction shall have the manufacturer's commercial, baked-on-epoxy finish.

3.6.10 Nameplate. Each sterilizer shall have one or more nameplates, permanently fastened (welded, bonded or with drive screws) and reasonably accessible, containing at least the name of manufacturer, manufacturer's type and model designation, serial number and electrical characteristics, contracts or purchase order number.

3.6.11 Workmanship. The products supplied under this specification shall be new and free from defects and imperfections that might affect their safety, serviceability, maintainability, and appearance.

3.6.12 Deilectric strength. Each sterilizer shall withstand the test specified in 4.2.3.3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspections.

4.2.1 Design and construction. Prior to shipment, each sterilizer shall be visually examined and operated to determine conformance to the design and construction requirements of this specification. Nonconforming sterilizers and parts will be rejected and replaced with conforming sterilizers or parts.

4.2.2 Materials and instruments. Physical and chemical tests of materials shall be conducted to determine conformance with these specifications. Nonconforming materials will be replaced with conforming materials. Certificates of quality from the supplier of materials and instruments (listed in 4.2.3.3.5, table VI) used in manufacturing the sterilizer, shall be acceptable in lieu of the sterilizer manufacturer performing such physical and chemical test to determine conformance to these specifications.

4.2.3 Performance. The supplier shall conduct performance inspections on each sterilizer covered by this specification, using the test methods specified herein. A sterilizer failing to pass the tests shall be replaced or reworked and retested until it satisfactorily passes the tests.

4.2.3.1 Test records. Records of tests satisfactorily performed on a sterilizer accepted by the Government under this specification shall be certified by the manufacturer and kept on file for two years, available for examination by the Government upon request. Such test records may be recertified once every 12 months, following their initial certification, for subsequent use as provided in 4.2.3.1.1.

4.2.3.1.1 Waivers of testing. The performance inspection tests specified in 4.2.3.3 through 4.2.3.3.6 may, at the procurement contracting officer's discretion, be waived for subsequent contracts which specify sterilizers conforming to those of the same type and size (other characteristics excluded) as previously tested, and for which the manufacturer has certified or recertified test records on file.

4.2.3.2.1 Apparatus and instruments. Apparatus and instruments used for testing the sterilizer shall be calibrated for accurate operation. The test instruments shall be calibrated at least annually to assure laboratory accuracy.

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4.2.3.2.2 Environment. Pressure and temperature come-up times shall be started with the sterilizer (and generator if a type II or IV) at ambient test area temperatures (see 4.2.3.3.1). Test loads shall be at ambient, test area temperature prior to conducting tests requiring them.

4.2.3.2.3 Installation and operation of sterilizer. Operation of the sterilizer during the tests shall be in accordance with the manufacturer's instructions specified in 3.5.3. Water for operation of the sterilizer shall be delivered at 20 to 50 psig dynamic; water to the generator of a type II or type IV sterilizer shall be not greater than 70°F (21°C) Steam shall be delivered to the sterilizer jacket at 50 to 80 psig dynamic.

4.2.3.3 Test methods.

4.3.3.3.1 Pressure and temperature come-up.

4.2.3.3.1.1 Generator pressure, types II and IV sterilizer:

Step 1 With the generator installed to the specified electric supply (see 6.2) and to line delivering water at 20 to 50 psig, 70°F (21°C).

Step 2 Close sterilizer steam port, actuate generator, and record the time required to reach 50 to 80 psig as shown on the generator pressure gauge.

The time required to reach generator pressure of 50-80 psig as shown on the 15 minutes maximum.

4.2.3.3.1.2 Sterilizer operational tests:

4.2.3.3.1.2.1 Manual control.

Step 1 Open steam to jacket valve, adjust pressure regulating valve until jacket pressure reads 15-17 psig. Open steam to chamber valve; measur and record the time required for chamber temperature to rise to 250° - 254°F as shown on temperature indicator; then operate sterilizer (with chamber at 250° - 254° for 30 minutes.

Step 2 Reduce chamber to ambient pressure.

Step 3 Adjust pressure regulating valve until jacket pressure gauge reads 27-30 psig. Open steam to chamber valve; measure and record the time required for chamber temperature to rise to 270° - 274°F as shown on temperature indicator; then operate sterilizer (with chamber at 270° - 274°F) for 30 minutes. Finally, reduce chamber to ambient pressure.

Step 4 Remove chart from recorder, check times and temperature with table IV below. Chamber come-up time and temperatures shall be within limits indi cated.

4.2.3.3.1.2.2 Automatically controlled sterilizers.

Step 1 Set temperature indicator-recorder-controller at 250°F., and timer for 30 minute sterilizing phase. Start sterilizer; measure and record time required for chamber temperature to rise to 250° - 254°F as shown on temperature indicator; then permit sterilizer to complete cycle.

Step 2 Reduce sterilizer chamber to ambient pressure.

Step 3 Set temperature indicator-recorder controller at 270°F and timer for 30-minute sterilizing phase. Start sterilizer; measure and record time required for chamber temperature to rise to 270° - 274°F as shown on the temperature indicator; then permit sterilizer to complete cycle. Finally reduce chamber to ambient pressure.

Step 4 Remove chart from recorder, check times and temperatures with table IV below. Chamber come-up time and temperatures shall be within limits indicated.

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TABLE IV

Sterilizer size	Chamber come-up (minutes)		Minimum Chamber temperature degrees F		Maximum Chamber temperature degrees F	
	250°F	270°F	250°	270°	250°	270°
1	4	5	248	268	256	276
2	4	6	248	268	256	276
3	6	7	248	268	256	276
4	7	8	248	268	256	276
5	10	12	248	268	256	276

4.2.3.3.2 Exhaust efficiency.

4.2.3.3.2.1 Fast exhaust. With no load in the chamber, operate the sterilizer to achieve chamber temperature of 270° - 274°F (131° - 134°C). Then actuate the fast-exhaust selector and measure the time required to reduce the chamber pressure to zero as shown on the chamber-vacuum-pressure gauge. The time required to reduce the sterilizing chamber pressure to zero shall not exceed three minutes.

4.2.3.3.2.2 Slow exhaust. This test shall be conducted as follows:

Step 1 Load sterilizing chamber with the quantity of bottles, with collars and caps in place, specified in table V, each containing 1,050 cc of water. (Bottles, collars and caps shall conform to DD-B-592). Suspend a thermocouple in the centermost bottle in the load and connect the thermocouple lead wire to a recording potentiometer.

Step 2 Actuate the potentiometer and operate the sterilizer to expose the load to 250° - 254°F (121 - 123°C) for 30 minutes, followed by slow exhaust of the chamber.

Step 3 Measure the time required for pressure in the sterilizing chamber to drop to zero, as indicated on the chamber-vacuum-pressure gauge.

The time required to return the chamber to ambient pressure shall be not greater than that specified in table V, with loss of not more than 5 percent of water from any bottle in the load, and without blowing any of the bottle stoppers or preventing automatic sealing of the bottles. The potentiometer shall have recorded 250° - 254°F (121 - 123°C) continuously for at least 12 minutes of the processing cycle.

TABLE V Solution exhaust times

Size	Number of bottles in load	Exhaust time (in minutes)
1	15	32
2	32	47
3	112	53
4	154	57
5	196	60

4.2.3.3.3 Sterilizing and drying efficiency. This test shall require a surgical-goods pack containing two paper strips, each inoculated with dried spore population of Bacillus stearothermophilus as specified in MIL-S-36586 adjusted for predetermined resistance as follows: to survive 250°F (121°C) for 5 minutes, to be killed within 13 minutes. The two spore-inoculated strips shall be placed in the center of the test pack which shall contain the following articles (freshly laundered and loosely wrapped in a double thickness of muslin): 3 surgical gowns, 12 towels, 30 gauze sponges (each 4 inches square), 5 lap sponges, and 1 drape sheet. The pack shall be approximately 12 inches by 12 inches by 20 inches and shall weigh 10 to 12 pounds.

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4.2.3.3.3.1 Procedure.

Step 1 Record the weight of the test pack and then place it (on edge) in the sterilizing chamber.

Step 2 Operate the sterilizer to expose the pack at 250°F for 30 minutes and dry it for 15 minutes (with sterilizer door closed pressure tight).

Step 3 Within 1 minute following completion of the drying phase of the cycle, remove the pack from the chamber and record its weight.

Step 4 Remove the spore-inoculated strips from the pack and test each strip for sterility as recommended in the U.S. Pharmacopoeia.

4.2.3.3.3.2 Results. Sterility of the test pack shall be evidenced by the killing of all spores on the spore-inoculated strips. Moisture retained by the pack shall cause not more than 3 percent increase in the weight of the pack prior to sterilizatio and the pack shall exhibit no wet spots.

4.2.3.3.4 Low-temperature system. A type III or type IV sterilizer equipped with a supplemental, low-temperature system (see 3.3.6.1.3) shall, in addition, be tested as follows: The tests shall be conducted with the sterilizer chamber containing 18 X 150-mm test tubes in quantities per Load A or 18 X 150-mm test tubes and 500-ml Erlenmeyer flasks in quantities per Load B. Test tubes shall contain 7 milliliters of water; flasks, 150 milliliters. Test tubes and flasks shall be capped but not sealed. Test tubes to be contained in wire baskets. The quantity of items in the load shall b as follows:

Size	Load A	Load B	
	No. of test tubes	No. of test tubes	No. of flasks
1	230	120	19
2	600	120	48
3	2700	120	108
4	3200	120	144
5	4000	120	180

4.2.3.3.4.1 Procedure:

Step 1 With selection of Load A distribute the baskets of test tubes in the sterilizer chamber. With selection of Load B place 40 test tubes in eac of 3, wire-form baskets and locate the baskets in the center, lower fron center and upper rear center of sterilizer chamber; distribute the flask in the sterilizer chamber.

Step 2 Insert thermocouples in test tubes (within the load) as follows: one nearest the center, one nearest the lower-front center, and one nearest the upper-rear center. Finally connect the thermocouples to a recording potentiometer.

Step 3 Close sterilizer door steamtight and operate the sterilizer to process the load at 78°C (172°F), for 45 minutes.

Step 4 Actuate the recording potentiometer.

Step 5 Following the test cycle, open the door and allow the load to cool to ambient temperature.

Step 6 Repeat steps 3, 4 and 5 but in steps 3 and 4 operate sterilizer to allow water in the test tubes to rise to 88°C (190°F).

Step 7 Repeat steps 3 and 4 but operate sterilizer to allow water in the test tubes to rise to 104°C (219°F).

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4.2.3.3.4.2 Allowable test-temperature variances. The temperature variances in the test tubes, as indicated by the potentiometric readings, shall not exceed the following during each of the 45-minute test processing-temeprature periods:

Processing Temperature (Degrees C/F)	Variance (Degree C/F - Plus or Minus)
78/172	3.0/6.0
88/190	2.0/4.0
104/219	1.0/2.0

4.2.3.3.5 Accurancy of instruments. When tested against certified laboratory standards, controls and instruments shall be accurate within the limits shown in table VI.

TABLE VI - Accuracy of instruments

Instrument	Test Range	Accuracy
Indicator-recorder-controller	160-280°F 72-138°C	± 2.0°F ± 1.0°C
Temperature-indicator-recorder	160-280°F	± 2.0°F, 1.0°C
Automatic sterilizing	0-90 min.	+ 4 min.
Exposure timer		- 0 min.(on repeated operation)
Mechanical exposure timer (on manually operated sterilizer)	0-60 min.	+ 6 min., - 0 min.
Drying timer (not integral with exposure timer)	0-60 min.	+ 4 min. - 0 min. (on repeated operation)
Vacuum-pressure gauge (chamber)	30 Hg-60 psig	± percent over total range
Jacket-pressure gauge	0-60 psig	± 1.5 psig

4.2.3.3.6 Temperature control. When measured by a thermocouple in the chamber drain line, sterilizing temperatures (between 212°F and 270°F) shall not fall below, nor exceed by 4°F, that set on the indicator-recorder-controller, or temperature indicator-recorder. If the sterilizer is equipped with a low-temperature system (see 3.6.3.1.3), operating temperature-stability of that system shall conform to the requirements of 4.2.3.3.4.2.

4.2.3.3.7 Dielectric strength. With the sterilizer chamber at the maximum temperature it is likely to encounter during normal operation, apply gradually increasing 60-Hz potential between the current-carrying parts and the metal enclosure. Voltage, as follows, shall be maintained for one minute:

Rate nominal voltage of component and circuitry	Test voltage to be applied
120 volts	900 volts
over 120 volts	1000 volts plus twice the nominal voltage

There shall be no arcing or other evidence of insulation or design failure.

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4.2.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table VII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defectives.

TABLE VII - Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion or container cushioning inadequate, improper, or missing.

4.2.4.1 Examination of closure and reinforcement of containers. When shipping containers are required to comply with PPP-B-601, PPP-B-621, or PPP-B-640, the examination for defects in closure and reinforcement shall be in accordance with the appendix of that specification.

4.2.4.2 A certificate of compliance shall be furnished with each sterilizer by the manufacturer, that the sterilizer supplied under this specification meets all the requirements as called for herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see section 6).

5.1.1 Level A.

5.1.1.1 Motors. Openings in electric motors shall be sealed with tape conforming to PPP-T-60, type II, class 1.

5.1.1.2 Indicators, gauges and dials. All indicators, gauges and dials shall be covered with cushioning material conforming to PPP-C-843 and the cushioning material secured in place with tape specified herein.

5.1.1.3 Switch boxes, outlets, connections and drain-line opening. Switch boxes, outlets, connections, and drain-line openings shall be sealed with tape specified herein.

5.1.1.4 Instruction books and parts lists. Instruction books and parts lists shall be packaged together in accordance with MIL-P-116, method IC-1, and secured to the sterilizer in a protected location.

5.1.1.5 Trays and racks. Trays and racks shall be secured by tying, blocking or bracing to prevent movement during transit.

5.1.2 Level B. The sterilizer shall be packaged as specified in 5.1.1.

5.1.3 Level C. The sterilizer shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see section 6).

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5.2.1 Level A.

5.2.1.1 Sterilizer. Each sterilizer shall be packed in container conforming to PPP-B-621, class 2, style optional: or to PPP-B-601, overseas type; or to PPP-B-640, class 2, grade A, style optional. When a sterilizer exceeds 1000 pounds net weight it shall be packed in a crate conforming to MIL-C-3774 nailed assembly, skid-type base. The contents of the crate shall be blocked, braced, and anchored in accordance with MIL-C-104, and waterproofed with a shroud extending to the base of the crate in accordance with the appendix to MIL-C-132. The contents of a nailed-wood or fiberboard container shall be waterproofed within a sealed-case liner conforming to MIL-L-10547, except that a sealed-case line may be omitted from a fiberboard container when it is sealed with water-resistant tape in accordance with the appendix to the fiberboard-container specification. Strapping shall be in accordance with the appendix to the applicable container specification.

5.2.1.2 Loading car. Each loading car shall be packed in a container conforming to PPP-B-621, class 2, style optional; or to PPP-B-601, overseas type; or to PPP-B-650, class 2, grade A, style optional. The contents of a nailed-wood or fiberboard container shall be waterproofed within a sealed-case liner conforming to MIL-L-10547, except that a sealed-case liner may be omitted from a fiberboard container when it is sealed with water-resistant tape in accordance with the appendix to the fiberboard-container specification.

5.2.2 Level B.

5.2.2.1 Sterilizer. Each sterilizer shall be packed as specified in 5.2.1.1 except that the container shall be class 1, domestic type and water-proofing shall not be required.

5.2.2.2 Loading car. Each loading car shall be packed as specified in 5.2.1.2 except that the container shall be class 1, domestic type; waterproofing shall not be required.

5.2.3 Level C. Each complete sterilizer shall be packed to ensure carrier acceptance and safe delivery to destination in container(s) complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, shipping container(s) shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, shipping container(s) shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The (gravity air removal type) steam sterilizers covered by this specification are only those types and sizes commonly supplied by the industry to fulfill ordinary hospital needs. Usually, sterilizers for large biomedical-research laboratories, hospital automated-materials-processing systems, and similar specialized applications, must be custom designed and equipped to best serve such needs on an individual basis. Consequently, when such special-purpose sterilizers are required, appropriate equipment specifications should be prepared locally in consultation with the concerned medical, architectural and equipment planning staffs and equipment manufacturers' representatives.

6.1.1 Materials-handling accessories. Special consideration should be accorded the options offered in table II (see 3.6.8) for materials-handling accessories. It is important that the sterilizers be ordered with the desired quantity of instrument trays, shelves or loading cars and carriages. The following should be considered when specifying loading cars and carriages.

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(a) Single-door sterilizer: Three sets, to provide one loading car in the sterilizer: Three sets, to provide one loading car in the sterilizer (with carriage standing by), one car and carriage at a work station, and one car and carriage in the post-sterilizing cooling area.

(b) Double-door sterilizer: Three sets for use as indicated in (a) above plus one additional carriage for the "unloading" end of the sterilizer.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, size, and style required (see 1.2).
(c) Number of copies and distribution of instruction books and parts lists, if different from requirements of 3.5.3 and 5.1.1.5; also additional information if required (see 3.5.3).
(d) Where instruction booklets and parts list are to be forwarded for review and approval (see 3.5.3.2).
(e) When the fifteen additional copies of data are required (see 3.5.3.3).
(f) State whether right- or left-hand door swing is desired (see 3.6.2).
(g) State whether power or manual operated door desired (see 3.6.2). NOTE: A power operated door shall be acceptable in lieu of a manual door when offered.
(h) Selection of automatic or manual controls (see 3.6.3).
(i) If a type III or type IV sterilizer, whether a supplemental, low-temperature control shall be furnished (see 3.6.3.1.3).
(j) Selection of condenser assembly or atmospheric exhaust (see 3.6.4).
(k) If sterilizer shall have a connection for piping the effluent condensate to waste (see 3.6.5.1 and 3.6.5.2.1).
(l) Electrical characteristics (208, 240, or 480 volts) (see 3.6.5.2.4).
(m) Selection of materials-handling accessories (see 3.6.8 and 6.1.1).
(n) Responsibility for inspection if different from the requirements of 4.1.
(o) Distribution of certified test records, if different from 4.2.3.1.
(p) Selection of applicable levels of packaging and packing (see 5.1).
(q) If for installation in the continental United States, supervision of installation, test and demonstration by a thoroughly qualified technician employed by the manufacturer and trained in his plant.
(r) State if canopy required (see 3.6.7.3 and 6.4).
(s) When interlock is required for a style C or D sterilizer (see 3.6.1.5 and 6.3).
(t) When biological seal is required for a style C or D sterilizer (see 3.6.4.3 and 6.3). If seal is required, also indicate its location, i.e., for partition wall at control end of sterilizer or at opposite end.

6.3 Interlock and biological seal. Either or both of these devices should be specified when highly infectious materials are to be loaded into a double-door sterilizer in one room be processed in the sterilizer and then be removed in an adjacent room. Interlocks and biological seals are normally specified only for sterilizers used in biomedical research laboratories and other special locations; therefore, before specifying these devices, the needs for them should be verified with the cognizant professional medical staff.

6.4 Canopy. When this accessory can be conveniently connected to the room air-exhaust system, sensible heat loss from the sterilizer to the room can be substantially reduced. This is because the room air can be allowed to flow through the louvered openings in the sterilizer cabinet panels and canopy overhang to entrain heat from the sterilizer body and moisture-laden air from the chamber (produced upon opening the door following a process cycle) and conduct them directly into the room air-exhaust system. Before specifying a canopy, the need for it should be verified by the hospital engineer.

Military Coordinating Activity:

DSA - DM

Civil Agency Coordinating Activities:

GSA - FSS
VA - MED

Preparing Activity:
VA - MED

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 45 cents each.

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